



National Institute of Technology Kurukshetra
Department of Computer Engineering
Digital System Design (CSIC100)
Mid Exam-I, FEB-2025

Timing: 50 Mins.

Max Marks: 15

All the Questions are Compulsory.

1. (a) The 16-bit 2's complement representation of an integer is 1111111111110101. Calculate its decimal representation. (1)
(b) Consider the equation $(123)_5 = (X8)_Y$ with X and Y as unknown. Calculate the possible solutions for X and Y. (1)
(c) Represent the decimal number 5137 in (a) BCD. (b) excess-3 code. and (c) 2421 code. (3)
2. (a) The representation of the value of a 16-bit unsigned integer X in Hexadecimal Number System is **BCA9**. Find the representation of X in Octal number system. (2)
(b) Perform the arithmetic operations $(-42) - (-13)$ in binary using signed-2's complement representation for negative numbers. Verify the answer. (2)
3. Assume that a 12-bit Hamming codeword consisting of 8-bit data and 4-check bits is $d_8d_7d_6d_5c_8d_4d_3d_2c_4d_1c_2c_1$ where the data bits and check bits are given in the following tables:
Find out the correct values of X and Y.

Data bits								Data bits			
d_8	d_7	d_6	d_5	d_4	d_3	d_2	d_1	c_8	c_4	c_2	c_1
1	1	0	X	0	1	0	1	Y	0	1	0

4. Show the step-by-step multiplication process using Booth's algorithm for $(+15) \times (-13)$. Assume 5-bit Registers that hold signed numbers. (3)